CFRM 502, Winter 2020.

Homework 6

Due: Monday 9th March 2020, NO LATER than 11:59pm

Instructions: You must submit your work through Canvas, and in the form of two attachments: A .pdf file of your detailed answers and a file containing your codes (.R or .Rmd). The .pdf files can be typeset or be a readable scan of your handwritten notes. You do not need to include R outputs in the .pdf file. However, you must clearly indicate the relevant part of your R codes when answering a question that is based on your code.

The maximum number of points you can receive for this homework is 50.

1. Consider the daily log return, in percentage, of Coca-Cola stock (KO) from January 3rd, 2007 to January 31st, 2018. You may obtain the data using "quantmod", or load it from the text file "HW6-Q1.txt" that is posted with this homework using the following code.

```
dat=read.table("HW6-Q1.txt", header=TRUE, sep = ",")
KO=xts(dat[,2], order.by=as.Date(dat[, 1]));names(KO) = c("KO")
r = 100*diff(log(KO))[-1]
```

   Let $\{r_t\}$ be the percentage daily log returns.

   (a) (2 points) Are there any autocorrelation in $\{r_t\}$? Are there any ARCH effect in $\{r_t\}$? Why?

   (b) (4 points) Build an ARMA+GARCH(1,1) model with normal residuals to $\{r_t\}$. Perform model checking and write down the fitted model.

   (c) (4 points) Build an ARMA+GARCH(1,1) model with Student-t residuals to $\{r_t\}$. Perform model checking and write down the fitted model.

   (d) (5 points) Build an ARMA+GARCH(1,1) model with skew-Student-t residuals to $\{r_t\}$. Perform model checking and write down the fitted model. Based on the model, are the returns of KO stock skewed?

   (e) (5 points) Does the return of KO stock has risk premium? You should answer based on an appropriate statistical model. Write down the model and perform model checking.

   (f) (5 points) Does the return of KO stock show leverage effect? You should answer based on an appropriate statistical model. Write down the model and perform model checking.

2. (a) (6 points) Consider the setting in Question 4 of Homework 5 (1 minute spot and futures log-prices of S&P 500). Build an ARIMAX+GARCH model for y with x as the predictor. Write down the fitted model. Perform Model checking. Resolve any issue that you encounter by extending the model. You only need to report your final model and show that it passes model checking.

   (b) (2 points) In the ARIMAX model in part (a), do negative shocks to the error term affect vol more than positive shocks? Your answer should be based on a statistical model. Write down the model and perform model checking.

3. Consider two NYSE traded stocks Billiton Ltd. (BHP) and Vale S.A. of Brazil (VALE). The daily adjusted closing price of the stocks, from July 2002 to March 2006, are included in the text file "HW6-Q3.txt", and can be accessed with the following code

```
dat=read.table("HW6-Q3.txt", header=TRUE, sep = ",")
VALE=xts(dat[,2], order.by=as.Date(dat[, 1]))
BHP=xts(dat[,3], order.by=as.Date(dat[, 1]))
names(VALE) = c("VALE");names(BHP) = c("BHP")
```

Let $\{p_{1,t}\}$ and $\{p_{2,t}\}$ be the daily log-prices of BHP and VALE, respectively.

(a) (2 points) Determine the integration order of the log prices.

(b) (3 points) Use the two-step method to determine if the log-prices are cointegrated. What is the cointegrating relationship? Denote the cointegrating relationship by $\{z_t\}$.

(c) (2 points) What is the mean of $\{z_t\}$? What is the standard deviation of $\{z_t\}$?

4. (a)(2 points) Consider the following error correction model:

$$\Delta \mathbf{Y}_t = \boldsymbol{\alpha}\boldsymbol{\beta}^\top \mathbf{Y}_{t-1} + \boldsymbol{\varepsilon}_t, \quad \{\boldsymbol{\varepsilon}_t\} \sim \text{ weak WN}$$

where $\{\mathbf{Y}_t\}$ are the observations, while $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are unknown matrices of parameters to estimated. Can $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ be uniquely estimated from the observations of $\{\mathbf{Y}_t\}$? Explain.

Consider 6 stocks with symbols GM, F, UTX, CAT, MRK, and PFE. Their daily adjusted closing prices are included in the text file "HW6-Q4.csv" and can be accessed with the following R code

```
Stock_FX_Bond = read.csv("HW6-Q4.csv", header=T)
adjClose = Stock_FX_Bond[,seq(from=3, to=13, by=2)]
```

Answer part (b), (c), (d) and (e) below by **only** using the output of the following code.

```
library(urca)
summary(ca.jo(adjClose))
```

(b)(2 points) Are the closing prices stationary?

(c)(2 points) How many cointegrating relationships exist among the stocks? Use 5% significance level, and explain your answer.

(d)(2 points) Write down the fitted error correction model.

(e)(2 points) Suggest a trading strategy based on the cointegration analysis.